

LM07 Analysis of Long-Term Assets

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1. Introduction

Long-term assets are defined as those assets that are expected to provide future economic benefits extending more than one year.

These assets include:

- **Tangible assets** also known as fixed assets or property, plant, and equipment. Examples include land, buildings, furniture, machinery, etc.
- **Intangible assets** lack physical substance. Examples include patents, trademarks, etc.

While a "economic" balance sheet would include a wide range of assets such as a company's reputation and its trained, experienced workforce, "accounting" balance sheets prepared in accordance with IFRS and US GAAP allow for the recognition of a narrow range of assets.

There are two important questions in accounting for a long-term asset:

- What cost should be shown on the balance sheet?
- How should this cost be allocated over the life of an asset?

2. Acquisition of Intangible Assets

Intangible assets lack physical substance. Classic examples include software, customer lists, patents, copyrights, and trademarks.

Under IFRS, identifiable intangible assets must meet three definitional criteria. They must be:

- Identifiable – either capable of being separated from the entity or arising from contractual or legal rights
- Under the control of the company
- Expected to generate future economic benefits

In addition, two recognition criteria must be met:

- It is probable that the expected future economic benefits of the asset will flow to the company.
- The cost of the asset can be reliably measured

Accounting for an intangible asset depends on how it is acquired.

Acquired in a Business Combination

This refers to a situation where one company buys another company and, in the process, acquires intangible assets.

- Both IFRS and US GAAP require the use of acquisition method in accounting for business combinations. (This method will be studied in detail at Level II.)
- Under the acquisition method, identifiable intangible assets such as patents, copyrights, and trademarks are recorded at their fair value.

- Goodwill is an intangible asset that cannot be identified separately. It is recorded when one business acquires another business. If the purchase price exceeds the fair value of the net identifiable assets (both the tangible assets and the identifiable intangible assets, minus liabilities) acquired. For example, Tan Hospitals Inc. acquires Man Equipments Inc. for \$100 million. The fair value of Man Equipments' net assets equal \$95 million. In this case, the excess of \$5 million will be recorded as goodwill.

Purchased in Situations Other than Business Combinations

This refers to a situation where an identifiable intangible asset is purchased, e.g. buying a patent from an inventor. The identifiable intangible asset is recorded at fair value which is assumed to be equal to the purchase price.

Developed Internally

Costs to internally develop intangible assets are generally expensed when incurred, although there are exceptions. The differences in whether the costs are capitalized or expensed affect financial statement ratios as outlined below:

- Balance sheet: A company that develops intangible assets internally will expense costs and record lower assets compared to a company that acquires such assets through purchase.
- Statement of cash flows: The costs of internally developing intangible assets are classified as operating cash flows, while the cost of acquiring intangible assets is classified as investing cash flows.

For internally developed intangible assets, there are two phases: the research phase and the development phase.

Research phase refers to the period during which commercial feasibility of an intangible asset is yet to be established. It is defined as "original and planned investigation undertaken with the prospect of gaining new scientific or technical knowledge and understanding."

Development phase refers to the period during which the technical feasibility of completing an intangible asset has been established with the intent of either using or selling the asset.

The treatment for the two phases varies slightly under IFRS and US GAAP as outlined below:

Under IFRS:

- Research costs are expensed.
- Development costs can be capitalized if technical feasibility and the intent to sell the asset are established.

Under US GAAP:

- Both research and development costs are expensed, but there are exceptions for software development.
- Software for sale: Costs incurred to develop a software product for sale are expensed

until the product's feasibility is established, and capitalized after the product's feasibility has been established. Determining feasibility involves judgment.

- Software for internal use: All development costs should be capitalized.

Example

Acme Inc. starts an internal software development project on January 1, 2012. It incurs expenditures of \$10,000 per month during the fiscal year ended December 31, 2012. By March 31, it is clear that the product will be developed successfully and will be used as intended. How are the software development costs recorded before and after March 31 according to IFRS and US GAAP?

Solution:

IFRS: Under IFRS all costs are expensed until feasibility is established if the software is developed for internal use. So, \$30,000 (period from January 1 to March 31, 2012) is expensed and \$90,000 is capitalized (from April 1 to December 31, 2012).

U.S GAAP: The entire cost of \$120,000 should be capitalized.

3. Impairment and Derecognition of Assets

Impairment of Assets

Impairment charges reflect an unexpected decline in the fair value of an asset to an amount lower than its carrying amount (Whereas depreciation and amortization charges allocate the cost of a long-term asset over its useful life.)

Under IFRS

- An asset is impaired when its carrying value exceeds the recoverable amount.
- The recoverable amount is the greater of (fair value less selling costs) and the (present value of expected cash flows from the asset, i.e., the value in use).
- If impaired, the asset is written down to the recoverable amount.
- Subsequent loss recoveries are allowed, but they cannot exceed the historical cost.

Under US GAAP,

- An asset is impaired if its carrying value is greater than the asset's undiscounted future cash flows.
- If impaired, the asset is written down to the fair value.
- Subsequent loss recoveries are not allowed.

Impact of Financial Statements

When an asset is impaired the impact in that period is:

- The value of the asset is written down.
- Activity ratios such as sales/assets are higher.

- Income is lower due to impairment expense.
- Therefore, profitability ratios are lower.
- Cash flows are not impacted (ignoring taxes).

The impact in subsequent periods is:

- Higher income because of reduced depreciation expense.
- Therefore, profitability ratios are higher.
- Activity ratios such as sales/assets are higher.

Example

Given the following data, what is the reported value under IFRS and US GAAP:

- Carrying amount = \$8,000
- Undiscounted expected future cash flows = \$9,000
- Present value of expected future cash flows = \$6,000
- Fair value if sold = \$7,000
- Costs to sell = \$200

Solution:

IFRS: Recoverable amount = greater of (\$7,000 - \$200, \$ 6,000) = \$6,800

Impairment loss = \$8,000- \$6,800 = \$1,200

Write down the value of asset from \$8,000 to \$6,800 in the balance sheet and record a loss of \$1,200 in the income statement.

US GAAP: Is the asset impaired? No, since the carrying amount of \$8,000 is less than the undiscounted future cash flows of \$9,000.

Other Impairment Scenarios

Impairment of Intangible Assets with Indefinite Lives: Intangible assets with indefinite lives are not amortized. They are carried on the balance sheet at historical cost, but they are tested annually for impairment.

Impairment of Long-term Assets Held for Sale: A long-term asset is reclassified as held for sale, if the management's intent is to sell it and its sale is highly probable. For example, if a company owns a machine with the intent of using it but now intends to sell it, then it should be reclassified as held for sale. Held-for-sale assets are not depreciated or amortized. At the time of reclassification, the asset should be tested for impairment and any impairment loss should be recognized.

The impairment loss can be reversed under IFRS and US. GAAP if the value of the asset recovers in the future. However, this reversal is limited to the original impairment loss. Therefore, the carrying value of the asset after reversal cannot exceed the carrying value before the impairment was recognized.

Derecognition of Assets

A company derecognizes an asset (i.e., removes it from the financial statements) when the asset is disposed of or is expected to provide no future benefits from either use or disposal.

The four ways in which an asset can be derecognized (removed from a company's financial statements) are as follows:

- *Selling the asset*: The difference between the sales proceeds and the carrying value of the asset is reported as a gain or loss on the income statement.
- *Abandoning the asset*: The carrying value of the asset is removed from the balance sheet and a loss is recognized in that amount in the income statement.
- *Exchanging the asset*: The carrying value of the old asset is compared to the fair value of the new asset and a gain or loss is reported.
- *Distributed to owners in a spin-off*: In a spin-off, typically, an entire cash generating unit of a company with all its assets is spun off and does not result in any gain or loss.

Impact on Financial Statements

A derecognition can result in either a gain or loss on the income statement.

- A loss will lead to lower net income and assets.
- A gain will lead to higher net income and assets.

4. Presentation and Disclosure

Under IFRS, for each class of property, plant, and equipment, a company must disclose the measurement basis, the depreciation method, the useful lives (or, equivalently, the depreciation rate) used, the gross carrying amount, the accumulated depreciation at the beginning and end of the period, and a reconciliation of the carrying amount at the beginning and end of the period.

Under U.S. GAAP, the requirements are less exhaustive. A company must disclose the depreciation expense for the period, the balances of major classes of depreciable assets, accumulated depreciation by major classes or in total, and a general description of the depreciation method(s) used in computing depreciation expense with respect to the major classes of depreciable assets.

The disclosures related to impairment losses also differ under IFRS and US GAAP. Under IFRS, a company must disclose for each class of assets the following:

- The amounts of impairment losses and reversals of impairment losses recognized in the period and where those are recognized on the financial statements.
- The main classes of assets affected by impairment losses and reversals of impairment losses and the main events and circumstances leading to recognition of these impairment losses and reversals of impairment losses.

Under US GAAP, reversal of impairment losses for assets held for use is not permitted. The

company must disclose the following:

- The description of the impaired asset, what led to the impairment;
- The method of determining fair value;
- The amount of the impairment loss, and where the loss is recognized on the financial statements.

5. Using Disclosures in Analysis

Ratios used to analyze fixed asset include: fixed asset turnover, asset age, and remaining useful life.

$$\text{Fixed asset turnover} = \frac{\text{Total revenue}}{\text{Average net fixed assets}}$$

The higher this ratio, the higher the amount of sales a company is able to generate with a given amount of investment in fixed assets. A higher asset turnover ratio is often interpreted as an indicator of greater efficiency.

$$\text{Average age of a company's assets} = \frac{\text{Accumulated depreciation}}{\text{Depreciation expense}}$$

$$\text{Average remaining life of a company's assets} = \frac{\text{Net PPE}}{\text{Depreciation expense}}$$

The older the assets and the shorter the remaining life, the more a company may need to reinvest to maintain productive capacity.

Assuming straight-line depreciation and no salvage value, we can state the following relationships:

$$\text{Estimated total useful life} = \text{Time elapsed since purchase (Age)} + \text{Estimated remaining life}$$

$$\text{Estimated total useful life} = \text{Historical cost} \div \text{annual depreciation expense}$$

$$\text{Historical cost} = \text{Accumulated depreciation} + \text{Net PPE}$$

The information presented in a company's disclosures and these relationships can be used to conduct an analysis on the company's long-lived assets.

Summary

LO: Compare the financial reporting of the following types of intangible assets: purchased, internally developed, and acquired in a business combination.

Purchased intangible assets

- The cost of a finite-lived intangible asset is amortized over its useful life.
- Indefinite-lived intangible assets are not amortized. They are tested for impairment at least annually.

Internally developed intangible assets

- Under IFRS, research costs are expensed but development costs may be capitalized.
- Under US GAAP, both research and development costs are expensed. (Except in the case of software created for sale to others.)

Intangible assets acquired in a business combination

- Acquired intangible assets such as patents, copyrights, and trademarks are recorded at their fair value; similar to long-lived tangible assets.

LO: Explain and evaluate how impairment and derecognition of property, plant, and equipment and intangible assets affect the financial statements and ratios.

Impairment

Under IFRS

- An asset is impaired when its carrying value exceeds the recoverable amount.
- The recoverable amount is the greater of (fair value less selling costs) and the present (value of expected cash flows from the asset i.e. the value in use).
- If impaired, the asset is written down to the recoverable amount.
- Subsequent loss recoveries are allowed, but they cannot exceed the historical cost.

Under US GAAP,

- An asset is impaired if its carrying value is greater than the asset's undiscounted future cash flows.
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Impact on financial statements

When an asset is impaired the impact in that period is:

- The value of the asset is written down.
- Activity ratios such as sales/assets are higher.
- Income is lower due to impairment expense.
- Therefore, profitability ratios are lower.

- Cash flows are not impacted (ignoring taxes).

The impact in subsequent periods is:

- Higher income because of reduced depreciation expense.
- Therefore, profitability ratios are higher.
- Activity ratios such as sales/assets are higher.

Derecognition

The three ways in which an asset can be derecognized (removed from a company's financial statements) are:

- Selling the asset: The difference between the sales proceeds and the carrying value of the asset is reported as a gain or loss on the income statement.
- Abandoning the asset: The carrying value of the asset is removed from the balance sheet and a loss is recognized in that amount in the income statement.
- Exchanging the asset: The carrying value of the old asset is compared to the fair value of the new asset and a gain or loss is reported.
- Distributed to owners in a spin-off: In a spin-off, typically, an entire cash generating unit of a company with all its assets is spun off and does not result in any gain or loss.

Impact on financial statements

- This can result in either a gain or loss on the income statement.
- A loss will lead to lower net income and assets.
- A gain will lead to higher net income and assets.

LO: Analyze and interpret financial statement disclosures regarding property, plant, and equipment and intangible assets.

An analyst can use financial statement disclosures to calculate the following:

Average age = accumulated depreciation/ annual depreciation expense.

Total useful life = historical cost/ annual depreciation expense.

Remaining useful life = ending PP&E/ annual depreciation expense.